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Mata kuliah : fuzzy logic
TTD :

1. Menggunakan Metode Mamdani

1.1. suhu tubuh 38.5°c (μ suhu normal (36-37.5))

fungsi linier turun : $\mu(x) = (b-x)/(b-a)$

karena $38.5 > 37.5$

μ_normal (38.5 = 0)

1.2. Suhu demam (37-39)

fungsi segitiga dgn Puncak di 38

~~Naik dari 37-38 : $\mu = (x-37)/(38-37)$~~ sisi kanan (38-39) : $\mu = (b-x)/(b/a)$

~~Turun dari 38-39~~ : μ_demam $(38.5) = (39-38.5)/(39-38) = 0.5/1 = 0.5$

1.3 Suhu tinggi (38-41)

μ_tinggi : fungsi naik

sisi naik : $\mu = (x-a)/(b-a)$

μ_tinggi $(38,5) = (38,5-38)/(39-38) = 0.5/1 = 0,5$

1.4 trombosit rendah

μ_rendah : fungsi turun (100-150)

$\mu = (b-x)/(b-a)$

$= (150-120)/(150-100) = 30/50 = 0,6$

1.5. trobosit normal

μ_normal : fungsi (100-150)

$\mu = (x-a)/(b-a)$

$= (120-100)/(150-100) = 20/50 = 0,4$

1.6. leukosit rendah

μ_rendah : fungsi turun (3-4)

$\mu = (b-x)/(b-a)$

$=$ ~~(4-3)/(4~~ $(4-3.8)/(4-3) = 0,2/1 = 0,2$

1.7 leukosit normal

μ_normal : fungsi naik (3-4)

$\mu = (x-a)/(b-a)$

$\mu = (3.8-3)/(4-3) = 0.8/1 = 0.8$

Hasil fuzzyfikasi

input	normal	Demam / rendah	Tinggi
suhu	0	0,5	0,5
trombosit	0,4	0,6	-
Leukosit	0,8	0,8	-

Evaluasi Rules dan operator min

Rule 1 : a = Min (0.5 , 0.6 , 0.2) = 0.2 → tinggi (S ∧ T ∧ L)

Rule 2 : a = Min (0.5 , 0.6) = 0.5 → tinggi (S ∧ T)

Rule 3 : a = Min (0.5 , 0.4) = 0.4 → sedang (S ∧ T)

Rule 4 : a = Min (0 , 0.4 , 0.8) = 0 → Rendah (S ∧ T ∧ L)

Rule 5 : a = Min (0.5 , 0.8) = 0.5 → sedang (S ∧ L)

Langkah 3. komposisi (Max)

μ rendah = Max (a_4) = Max (0) = 0

μ sedang = Max (a_3, a_5) = Max (0,4 , 0.5) = 0.5

μ tinggi = Max (a_1 , a_2) = Max (0.2 , 0,5) = 0,5

Langkah 4 : Defuzzyfikasi (centroid)

$$z^* = \frac{\int \mu(z) \times z \times dz}{\int \mu(z) \times dz}$$

$$z^* = \frac{\Sigma(\mu_i \times z_i \times A_i)}{\Sigma(\mu_i \times A_i)}$$

Perhitungan

sedang [30-70] , μ = 0.5

titik tengah : 50

lebar : 20

luas : μ × lebar = 0.5 × 20 = 10

momen : μ × z × lebar = 0.5 × 50 × 20 = 500

tinggi (60 – 100) ; μ = 0.5

titik tengah : 85

lebar : 30

luas : μ × lebar : 0.5 × 30 = 15

momen : μ × z × lebar : 0.5 × 85 × 30 = 1275

Hasil Akhir

z^* = (500 + 1275) / (10 + 15)

z^* = 1775 / 25

z^* = 71

Hasil Resiko DBD = 71 / 100 (tinggi)

2 input : $x_1 = 60$ (soil_moisture %), $x_2 = 26$ (soil_PH)
mode sugeno orde -1,
$f_i = p_i \cdot x_1 + q_i \cdot x_2 + r_i$

Rule	p	q	r
1	0.0692	0.0961	0.00231
2	0.0698	0.0390	0.00139
3	0.0585	0.0183	0.00084
4	0.0505	0.0318	0.00115

Perhitungan manual Anfis
1. fuzzyfikasi
fungsi Gaussian
$\mu(x) = \exp(-((x-c)^2)/(2\sigma^2))$
Asumsi A_1 ($c=40$, $\sigma=15$) A_2 ($c=70$, $\sigma=15$), B_1 ($c=20$, $\sigma=8$), B_2 ($c=30$, $\sigma=8$)
untuk $x_1 = 60$,
$\mu\_A_1(60) = \exp(-((60-40)^2)/(2 \times 15^2))$
$= \exp(-(400)/(450))$
$= \exp(-0.889) = 0.4111$
$\mu\_A_2(60) = \exp(-((60-70)^2)/(2 \times 15^2))$
$= \exp(-(100)/(450))$
$= \exp(-0.222) = 0.8007$
untuk $x_2 = 26$:
$\mu\_B_1(26) = \exp(-((26-20)^2)/(2 \times 8^2))$
$= \exp(-(36)/(128))$
$= \exp(-0.281) = 0.7548$
$\mu\_B_2(26) = \exp(-((26-30)^2)/(2 \times 8^2))$
$= \exp(-(16)/(128))$
$= \exp(-0.125) = 0.8825$
firing strength
$w_1 = \mu\_A_1 \times \mu\_B_1 = 0.4111 \times 0.7598 = 0.3103$
$w_2 = \mu\_A_1 \times \mu\_B_2 = 0.4111 \times 0.8825 = 0.3628$
$w_3 = \mu\_A_2 \times \mu\_B_1 = 0.8007 \times 0.7598 = 0.6099$
$w_4 = \mu\_A_2 \times \mu\_B_2 = 0.8007 \times 0.8825 = 0.7066$
total $w = 0.3103 + 0.3628 + 0.6099 + 0.7066 = 1.9891$

Normalisasi

$\bar{w} = wi / \Sigma w$

$\bar{w}_1 = 0.3103 / 1.9891 = 0.1569$

$\bar{w}_2 = 0.3628 / 1.9891 = 0.1828$

$\bar{w}_3 = 0.6099 / 1.9891 = 0.3096$

$\bar{w}_4 = 0.7066 / 1.9891 = 0.3561$

verifikasi $\Sigma \bar{w} = 0.9999 \approx 1.0$

output tiap rule

Rule 1 , $f_1 = 0.0692 \times 60 + 0.0461 \times 26 + 0.00231$

$= 4.152 + 1.1986 + 0.00231$

$= 5.3529$

Rule 2 , $f_2 = 0.0698 \times 60 + 0.0390 \times 26 + 0.00139$

$= 4.188 + 1.014 + 0.00139$

$= 5.2034$

Rule 3 , $f_3 = 0.0585 \times 60 + 0.0183 \times 26 + 0.00084$

$= 3.51 + 0.4758 + 0.00084$

$= 3.9866$

Rule 4 , $f_4 = 0.0505 \times 60 + 0.0318 \times 26 + 0.00115$

$= 3.03 + 0.8268 + 0.00115$

$= 3.8580$

Agregasi : $f = \Sigma (\bar{w}i \times fi)$

$f = (0,1569 \times 5,3529) + (0,1828 \times 5,2034) + (0,3096 \times 3,9866) + (0,3561 \times 3,8580)$

$f = 0,8372 + 0,9512 + 1,2147 + 1,3741$

$f = 4,3772$ ton / ha hasil produksi jagung